

STAT 240 Midterm 1 cheatsheet

GGplot functions:

Function Name	Description	Syntax
ggplot()	Create a new ggplot	Ggplot(data, mapping,...)
aes()	Defines variable aesthetics	aes(x = ..., y = ..., ...)
geom_point()	Creates a scatterplot	geom_point(mapping, col,...)
geom_line()	Creates a line graoh	geom_line(mapping, col,...)
geom_smooth()	Creates a smooth trendline	geom_smooth(mapping, method,...)
geom_col()	Creates a bar graph with x and y variable	geom_col(mapping, col,...)
geom_bar()	Creates a bar graph with a variable	geom_bar(mapping, fill, ...)
geom_histogram()	Creates a histogram for a variable	geom_histogram(mapping, fill, ...)
geom_density()	Creates a density plot for a variable	geom_density(mapping, fill, ...)
geom_boxplot()	Creates a boxplot for a variable	geom_boxplot(mapping, fill, ...)
labs()	Creates labels for a ggplot	labs(title, subtitle, ...)

Dplyr functions:

Function name	Description	Syntax
<code>mutate()</code>	Create, modify, and delete columns	<code>mutate(data, col1 = vector1, ...)</code>
<code>relocate()</code>	Change column order	<code>relocate(data, col1, col2, ...)</code>
<code>rename()</code>	Rename columns	<code>rename(data, new_name = old_name,...)</code>
<code>select()</code>	Keep or drop columns using their names	<code>select(data, list of column names)</code>
<code>arrange()</code>	Order rows using a column's values	<code>arrange(data, column name)</code>
<code>desc()</code>	Sort in descending order	<code>desc(column name)</code>
<code>filter()</code>	Keep or drop rows that match a condition	<code>filter(data, logical statements dealing with one or more columns)</code>
<code>slice_min()/</code> <code>slice_max()</code>	Subset rows using the mins/maxs of a given column	<code>slice_...(data, column name, n = ...)</code>
<code>summarize()</code>	Summarizes the dataset or each group down to one row	<code>summarize(data, col_name = summary function, ...)</code>
<code>group_by()</code>	Create groups inside one or more columns	<code>group_by(data, column names)</code>
<code>n()</code>	No. of rows in the "current" group or variable	<code>n()</code>
<code>count()</code>	Count the observations in each group	<code>count(column names)</code>

<code>case_when()</code>	Creates customized groups in a separate column	<code>case_when(logical statements ~ label of group)</code>
<code>left_join()</code>	A mutating join that keeps all observations in A	<code>left_join(A, B, by = ...)</code>
<code>right_join()</code>	A mutating join that keeps all observations in B	<code>right_join(A, B, by = ...)</code>
<code>inner_join()</code>	A mutating join that keeps matching observations from x and y	<code>inner_join(A, B, by = ...)</code>
<code>full_join()</code>	A mutating join that keeps all observations in x and y.	<code>full_join(A, B, by = ...)</code>
<code>semi_join()</code>	A filtering join that returns all rows from x with a match in y	<code>semi_join(A, B, by = ...)</code>
<code>anti_join()</code>	A filtering join that returns all rows from x without a match in y	<code>anti_join(A, B, by = ...)</code>

Tidy-R functions:

Function name	Description	Syntax
<code>pivot_longer()</code>	Pivot data from wide to long	<code>pivot_longer(data, vector of column names, names_to = ..., values_to = ..., ...)</code>
<code>pivot_wider()</code>	Pivot data from long to wide	<code>pivot_wider(data, names_from = ..., values_from = ..., ...)</code>
<code>drop_na()</code>	Drop rows containing missing values	<code>drop_na(data, ...)</code>